Multidecadal weakening of East India Coastal Current: Evidence from coastal altimetry in Northern Indian Ocean.

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Abstract

The East India Coastal Current (EICC) plays a crucial role in the coastal dynamics of the Bay of Bengal, yet its long-term variability remains difficult to quantify due to the limitations of traditional satellite altimetry near land. To overcome these observational challenges, this study utilizes the high-resolution X-TRACK (L2P v2.2) reprocessed along-track Sea Level Anomaly (SLA) dataset for the Northern Indian Ocean. The X-TRACK algorithm significantly improves the recovery of valid sea level observations in the coastal zone compared to standard products, enabling a more accurate assessment of near-shore currents. By analyzing this combined multi-mission altimetry record, the study identifies a multidecadal weakening trend in the EICC. This long-term weakening is further evaluated in relation to regional atmospheric forcing, specifically variations in wind stress magnitude over the Bay of Bengal gyre and the equatorial band. Ultimately, the findings provide critical observational evidence of long-term changes in boundary current dynamics in the Northern Indian Ocean.

1. *Introduction*

Reliable estimation of coastal currents is of great importance owing to their utility in several coastal services. Despite their importance, the in situ observations of currents along the coast are sparse and model currents suffer limitations with spatial resolution and boundary currents. Approximate geostrophic currents can be estimated with traditional altimetry products, but these have several limitations, especially on the coast. Most of the altimetry products are masked near coast due to reasons like signal contamination by land, inland water reflections which affect the shape of the reflected waveform. In addition to this, tides from global tidal models are inaccurate near the coast and shelf regions and reference Mean Sea Surface(MSS) is not accurate in the coastal regions due to lack of sufficiently long data in the coastal regions. Atmospheric effects are not properly accounted for while deducing sea level near the coast. The gridding procedure also smooths out the variability near the coast.

Coastal Altimetry algorithms like XTRACK (Roblou, 2007) aims to address several of the above challenges with various techniques. State of the art high resolution regional models are used to better filter out tides and meteorologically induced signals. Employing new waveform models for retracking of coastal waveforms (ALES, i.e., Advanced Leading Edge Subwaveform retracker) (Passaro, 2014) reduces the number of points rejected near the coast. Sophisticated vertical reference frame with updated MSS and Screening multiple along track altimetry data together are other techniques. Using these methods, many of the near coast points can be retained which would otherwise be edited out.

Before the advent of coastal altimetry, coastal currents and sea level were studied using regular satellite altimetry. Strub, 1997 for example, have examined the dynamics of the California Current System using TOPEX altimetry derived surface currents demonstrating the potential for studying large-scale circulations. One major application of coastal altimetry is to study sea level variability very close to the coast. One of the earliest studies using specialized coastal altimeter processing (Vignudelli et al., 2005) have studied sea level variability in Corsica Channel. They observed that open ocean tide gauge site Capraia has shown lesser sea level RMSE in comparison to near coast tide gauge site Livorno. This could be due to generally noisier radar reflections near the coast than in the open ocean. Specifically in North Indian Ocean context, Durand, 2009 used newly re-processed altimetry(MAP group) data of Topex/poseidon (T/P) to study the spatiotemporal variations of the East India Coastal Current (EICC). They were able to look into the cross shore structure of the EICC and showed the boundary trapped nature of EICC extending up to 150-200km. Dohan and Maximenko (2010) highlight the

importance of satellite altimetry and ocean vector winds for monitoring geostrophic and Ekman currents, providing insights into surface current dynamics at various scales.

Reprocessed coastal sea level data has been used to study several regional coastal circulations (Le Hénaff, 2011; Liu, 2012). While assessing the performance of XTRACK altimetry derived currents in the West Florida Shelf (Liu et al., 2012) have found that Ekman corrected geostrophic currents improve RMSE when compared with ADCP and HF radar observations.

Jebri, 2016 have shown that close to the coast in Mediterranean Sea, higher variability (in terms of standard deviation) is observed in XTRACK SLA compared to AVISO along-track SLA. They have also shown that XTRACK derived cross-track geostrophic currents are in better agreement with nearby ADCP than with AVISO derived currents. The analysis over South Atlantic Bight (Yuan, 2017) highlights the significant contribution of wind-driven Ekman dynamics to along-shelf transport, especially during seasons when wind is strong.

Since the above studies, several new developments have happened in coastal altimetry. X-TRACK v.2.7.12 version has updated Mean Dynamic Topography estimated from a combination of altimetry, gravimetry (GRACE and GOCE) and in-situ data, updated tides based on FES2014b model. A full description of the XTRACK algorithm can be found in (Birol et al., 2017). Very recently, Cazenave et al., 2022 demonstrated that reprocessed satellite altimetry data can be used to accurately capture coastal sea level trends within 20km of the shoreline. Their analysis indicates that coastal altimetry data can be used as virtual tide gauge stations to monitor coastal sea level rise across the globe including North Indian Ocean.

However, earlier coastal altimetry studies in the region, such as Durand et al. (2009), largely considered Ekman components to be negligible, focusing primarily on the geostrophic flow. To accurately capture the total surface current variability, especially in a monsoon-dominated regime like the Bay of Bengal where wind forcing is highly seasonal and intense, the integration of Ekman velocities with geostrophic currents is essential (Sudre et al., 2013).

Furthermore, while the intraseasonal and seasonal variability of the East India Coastal Current (EICC) has been extensively documented through hydrographic surveys, short-term mooring deployments (Mukherjee et al., 2014; Amol et al., 2014), and numerical modeling (McCreary et al., 1996; Shankar et al., 1996), its multidecadal evolution remains poorly understood.

Standard gridded altimetry products systematically fail to resolve this narrow, boundary-trapped flow over the continental slope. This observational gap is particularly concerning given the rapid and asymmetric warming of the Indian

Ocean, which has been shown to dampen overall monsoonal wind circulation and alter large-scale atmospheric patterns (Roxy et al., 2015; Schott et al., 2009). Consequently, there is a critical need to assess how boundary currents like the EICC are responding to sustained atmospheric and thermodynamic shifts over the basin.

To bridge this gap, this study leverages a reprocessed, multi-mission coastal altimetry dataset (X-TRACK) spanning over three decades, explicitly combined with Ekman transport estimates, to provide a comprehensive analysis of the EICC right up to the continental slope. By applying specialized coastal processing algorithms, the X-TRACK dataset recovers significantly more valid sea level anomaly (SLA) observations near the coast compared to standard Level-2 products (Figure 1), enabling the continuous observation of near-shore dynamics. With some missions (see Sup Fig 1) of XTRACK product spanning more than three decades at 10 days resolution, the dataset offers an unprecedented opportunity to study long term changes in the coastal sea level and currents in the Indian Ocean.

Our objective is twofold: first, to evaluate the performance of the XTRACK coastal altimeter product in the Northern Indian Ocean, by comparing XTRACK-derived geostrophic current components with data from HF radar, RAMA, OMNI buoys and drifter buoys. Special emphasis is given to estimate the effect of Ekman correction on the total velocity estimation in the nearshore regions and second, to investigate a sustained multidecadal weakening of the EICC. We explore the dynamic links between this weakening boundary current and the basin-wide relaxation of its primary drivers namely, local alongshore winds and remote equatorial wind stress, which modulate the propagation of coastal Kelvin waves.

The remainder of the paper is organized as follows: Section 2 details the multi-mission satellite altimetry data, wind products, and the methodology used to compute the total surface currents. Section 3 presents the validation of the altimetry-derived currents against in situ observations. Section 4 examines the multidecadal variability of the EICC and characterizes the dynamic response to shifting regional wind forcing.

2. ***Data and Methods***

The along-track Sea Level Anomaly (SLA) data processed with XTRACK (L2P v2.2) are grouped by regions and satellite missions that have similar orbital paths. For the current study, data for the northern Indian region (labeled as NINDIAN which covers 42E:99E, 0:25N) was downloaded from the ftp site ftp-access.aviso.altimetry.fr distributed by Center de Topographie des Océans et de l′Hydrosphére (CTOH). Supplemental Figure 1a shows the multi-mission reference tracks, their temporal and spatial resolution, total number of tracks (and their intersection points over the basin) available for the region NINDIAN. The horizontal

extent of the bars indicates the temporal coverage of the satellite missions. The temporal resolution ranges from 10 days for TOPEX/Poseidon (T/P) and Jason satellites to about one month for Sentinel-3A, ERS, Envisat and Saral combination. The along-track data has a spatial resolution of 6-7km corresponding to 1Hz sampling. The multi-mission references tracks also have varying levels of data availability (Sup Figure 1b). Missions S3A, TP+J1+J2+J3+S6A and S3B have the best overall data availability followed by HY2A, GFO and ERS1+ERS2+ENV+SRL.The multi-mission reference track TPN+J1N+J2N+J3N has both poor data availability and least number of intersection points covering the region.

To evaluate the computed geostrophic currents in the open ocean surface current data of RAMA buoys available at https://www.pmel.noaa.gov/tao/drupal/disdel/ and OMNI buoy current data has been used. To compute region averaged winstress(and curl) and Ekman velocity component, Cross Calibrated Multi-Platform (CCMP) v3.1 wind data available on 0.25° spatial grid has been used. This dataset is available at www.remss.com with DOI of https://doi.org/10.56236/RSS-uv6h30. Near shore currents are validated with the HF radar data from three locations Tamil Nadu (TN), Andhra Pradesh (AP) and Odisha (OR) on the east coast of India. These radar stations are operated and maintained by NIOT as part of Indian Coastal Ocean Radar Network (ICORN). Drifter buoy data from the Global Drifter Program (GDP) of the NOAA Atlantic Oceanographic and Meteorological Laboratory (AOML) has been used to further verify the robustness of computed geostrophic currents especially in the coastal regions. The data can be found at https://www.aoml.noaa.gov/ftp/pub/phod/buoydata/6h/. The dataset which maps the distance to the nearest coast is used to estimate proximity of observations to the coast. This dataset is available as part of Generic Mapping Tools (GMT) which is available at https://oceancolor.gsfc.nasa.gov/resources/docs/distfromcoast/.

To evaluate the performance and added value of the X-TRACK coastal altimetry product, standard along-track Level-2+ (L2P) Sea Level Anomaly (SLA) data distributed by AVISO+/CNES were utilized as a baseline for comparison. The L2P dataset provides global, 1-Hz along-track SLA time series derived from standard Level-2 instrumental measurements. In this standard product, SLAs are computed using state-of-the-art environmental and geophysical corrections (including wet/dry tropospheric, ionospheric, tidal, and dynamic atmospheric corrections) optimized primarily for the open ocean.

While highly accurate in the open ocean, standard L2P processing applies rigorous data-editing criteria that frequently flag and discard measurements near the coast. In this study, the standard deviation of SLA from the conventional L2P product was compared with that of the X-TRACK product as a function of distance from the coast. Because the X-TRACK processing chain uses the L2P dataset as its primary

input but applies specialized, coastal-oriented spatial filtering and editing strategies, comparing the two directly isolates the ability of the X-TRACK algorithm to recover valid, high-resolution physical variability in the near-shore region that would otherwise be lost or smoothed out in the standard L2P product.

In the XTRACK data product, the satellite ground tracks data were arranged with their first index falling on equator and last index falling on the coast. Variables such as SLA and Dynamic Atmospheric Correction (DAC), among others, are functions of points_numbers (spatial sampling points) and cycles_numbers(temporal). To better manipulate and visualize the data, the variables are reshaped as a function of distance (s) from the coast and time. The distance (s) is the along-track distance between coastal point and track points on an ellipsoidal model of earth computed based on geodesic algorithms(Karney, 2013). Raw data can give large gradients upon differentiation along track and then give unrealistic currents. Proper smoothing is needed to derive reliable currents.

An exercise has been done to compare the performances of various filters, namely, Gaussian with sigma levels of 3 and 6, box smoothing with window lengths of 6 and 12, optimal difference operator(Powell and Leben, 2004) and loess filtering in properly estimating the geostrophic current components at the track intersection points. Among the tested smoothing techniques, the along-track smooth by loess filter (Cleveland 1979) with 0.05 fraction of regional points for the regression is found to be best for estimating geostrophic currents. This filter gives an RMSE of 0.133m/s and correlation of 0.66 for computed geostrophic zonal currents against AVISO msla currents(extracted at intersection points) along the time axis. Optimal difference operator is second best with correlation, RMSE pair of 0.65 and 0.135. These are followed by other techniques. Hence, each track is smoothed using a loess filter with a fraction of 0.05 of data selected at each point for local regression. Meridional currents are poorly resolved with correlations <0.4 even with best smoothing filters. This is because all tracks make small acute angles with the north meridian at the intersection points. Hence most of the analysis is limited to well resolved zonal component of the current. Since geostrophic currents are undefined at equator, tracks within the band 5S-5N are trimmed. Differentiation along the track gives geostrophic current that is normal to the path of the track. The cross-track geostrophic current(*gc*) is calculated as follows,

$$gc = \frac{-g}{f}\frac{\partial h}{\partial s}$$

where *s* is the distance from coast which serves as along track dimension, *h* is the sea level anomaly, g is the acceleration due to gravity and f is the Coriolis parameter. For more precision, g is computed as a function of latitude.

To be able to resolve zonal and meridional components of the current vector, we need two cross-track geostrophic currents. This is only possible at the intersection points of ascending and descending tracks. Cross-track geostrophic currents (gc_1 and gc_2) for each track at the intersection points are computed. These are expressed as the linear combination of components of the current vector (Parke et al., 1987).

$$gc_1 = -usin(\alpha_1) + vcos(\alpha_1)$$
$$gc_2 = -usin(\alpha_2) + vcos(\alpha_2)$$

Here α_1 and α_2 are the heading angles (counter clockwise to the positive x-axis) of the intersecting tracks. We can deduce current vector components as follows.

$$u = \frac{gc_1 \cos\alpha_2 - gc_2 \cos\alpha_1}{\sin(\alpha_2 - \alpha_1)}$$
$$v = \frac{gc_2 \sin\alpha_1 - gc_1 \sin\alpha_2}{\sin(\alpha_2 - \alpha_1)}$$

To estimate the wind-driven Ekman current components, daily 10-meter wind velocity fields (u_{10}, v_{10}) were obtained from the Cross-Calibrated Multi-Platform (CCMP) version 3.1 dataset. The surface wind stress vector (τ_x, τ_y) was first calculated using the bulk aerodynamic parametrization:

$$\tau_x = \rho_{air} C_d W u_{10}$$

$$\tau_y = \rho_{air} C_d W v_{10}$$

where W is the wind speed magnitude ($\sqrt{u_{10}^2 + v_{10}^2}$), $\rho_{air} = 1.225\,\mathrm{kg\,m^{-3}}$ is the density of air, and $C_d = 1.5 \times 10^{-3}$ is the dimensionless drag coefficient. The kinematic wind stress components ($\tilde{\tau}_x, \tilde{\tau}_y$) were subsequently obtained by dividing the wind stress by the reference density of seawater ($\rho_{water} = 1025.0\,\mathrm{kg\,m^{-3}}$). Following classical Ekman theory (Vallis, 2006), the depth-dependent Ekman velocity components ($u_E(z), v_E(z)$) were computed using the analytical solutions for a wind-driven surface boundary layer in a homogeneous ocean of infinite depth. The equations governing the horizontal Ekman spiral are given by:

$$u_E(z) = \frac{\sqrt{2}}{f\,d} e^{z/d} \left[\tilde{\tau}_x \cos\left(\frac{z}{d} - \frac{\pi}{4}\right) - \tilde{\tau}_y \sin\left(\frac{z}{d} - \frac{\pi}{4}\right) \right)$$

$$v_E(z) = \frac{\sqrt{2}}{f\,d} e^{z/d} \left[\tilde{\tau}_x \sin\left(\frac{z}{d} - \frac{\pi}{4}\right) + \tilde{\tau}_y \cos\left(\frac{z}{d} - \frac{\pi}{4}\right) \right)$$

where z is the depth (defined as negative downwards, with $z = 0$ at the surface), $f = 2\Omega \sin(\theta)$ is the Coriolis parameter at latitude θ, and d is the Ekman layer depth. The Ekman depth is defined as:

$$d = \sqrt{\frac{2A}{|f)}}$$

where A represents the vertical eddy viscosity parameter, which was set to a constant value of $0.1\,\mathrm{m^2 s^{-1}}$. Because the Coriolis parameter approaches zero at the equator, causing singularities in the Ekman depth and velocity calculations, the computations were masked out for the equatorial band between 5°S and 5°N.

The momentum transferred from the wind to the ocean surface is highly sensitive to wind speed due to varying surface roughness. Following the piecewise formulation utilized by Wang et al. (2023), which builds upon the foundational bulk parameterizations of Large and Pond (1981), the dimensionless drag coefficient (C_d) was parameterized as a function of the 10-meter wind speed magnitude (W):

$$C_d = \begin{cases} (0.62 + 1.56/W) \times 10^{-3} & \text{if } W < 3.0\,\mathrm{m\,s^{-1}} \\ 1.14 \times 10^{-3} & \text{if } 3.0 \le W \le 10.0\,\mathrm{m\,s^{-1}} \\ (0.49 + 0.065\,W) \times 10^{-3} & \text{if } W > 10.0\,\mathrm{m\,s^{-1}} \end{cases}$$

Similarly, the assumption of a constant vertical eddy viscosity ($A = 0.1\,\mathrm{m^2 s^{-1}}$) limits the model's ability to accurately capture the depth of the Ekman layer, which deepens under stronger wind forcing due to enhanced turbulent mixing. To address this, a dynamic, wind-dependent eddy viscosity was implemented. Adapting the relationships established by Santiago-Mandujano and Firing (1990) and subsequently validated in the context of global ocean surface current products (OSCAR) by Johnson et al. (2007), A is defined dynamically as:

$$A = \begin{cases} a & \text{if } W < 1.0\,\mathrm{m\,s^{-1}} \\ a\,W^b & \text{if } W \ge 1.0\,\mathrm{m\,s^{-1}} \end{cases}$$

where the empirical constants are set to $a = 8 \times 10^{-5}$ and $b = 2.2$.The incorporation of these dynamic variables allows the Ekman layer depth ($d = \sqrt{2A/|f)}$) to adjust realistically to local atmospheric conditions.

RAMA, OMNI and radar daily currents are smoothed with a 72 hour low pass butterworth filter (4th order) to retain only sub-inertial currents. The 6 hourly drifter buoy data is averaged to daily and smoothed over a 3 day running window.

Intersection points of tracks don't usually fall on the points of observation to compare with. So, for each point observation we want to compare with, nearby intersection points are searched. Only those combinations of tracks and observations with less than a minimum threshold (0.5°) of distance are selected for comparison.

3. Results

The efficacy of the coastal-dedicated X-TRACK processing algorithm against standard AVISO Level-2+ (L2P) products is demonstrated in Fig 1. Figure 1a

illustrates the spatial distribution of 12 selected Sentinel-3A (S3A) satellite ground tracks over the Bay of Bengal. Historically, standard satellite altimetry has exhibited significant limitations in the coastal zone typically within 50 km of the shoreline. These limitations primarily stem from the contamination of the radar footprint by landmasses, inaccurate coastal tidal models, and degraded wet tropospheric corrections, which collectively corrupt the returning waveform and subsequent SLA estimates (Bouffard et al., 2010; Benveniste et al., 2020).

Standard processing protocols often result in a severe loss of data near the coast due to strict editing criteria that flag land-contaminated observations as invalid. Figure 1b shows the percentage of valid SLA observations recovered by both products. While the availability of standard L2P data drops precipitously as the satellite ground track approaches the coast, X-TRACK recovers a substantially higher percentage of valid points right up to the shoreline. By recovering these near-shore observations without amplifying the noise, X-TRACK effectively bridges the observational gap between the open ocean and the coastal domain. This recovery is critical for resolving narrow, coastal-trapped boundary currents such as the East India Coastal Current (EICC), which often occupy the heavily masked 0–50 km coastal band (Durand et al., 2009). Figure 1c illustrates the temporal standard deviation (STD) of the SLA as a function of distance from the coast. In the standard L2P product, the SLA variance increases sharply near the coastline. It can be seen that the XTRACK-processed data retains higher SLA variability near the coast (within the first 50 km) compared to the standard L2P product. Beyond approximately 150 km offshore, the standard deviations of both products converge. This improvement aligns with previous regional studies (e.g., Roblou et al., 2010; Jebri et al., 2016), which demonstrated that applying tailored along-track editing criteria and improved high-resolution geophysical corrections substantially reduces non-physical variance in coastal sea level time series.

3.1 Open ocean validations with RAMA/OMNI

Following the computation of the geostrophic and Ekman velocity components at the intersection points, it is crucial to validate the combined surface current estimates against independent in situ observations to test robustness of geostrophic current computation and other processing methods involved. For this purpose, we first utilized open-ocean near-surface current measurements from the OMNI(and RAMA) buoy network in the Bay of Bengal(BoB) and Arabian Sea(AS). Because of the temporal gaps in OMNI data and altimetry data, having a complete timeseries of overlapping periods is hard to find. Figure 2a illustrates one such spatial configuration used for the comparison. The black marker denotes the intersection point of two Sentinel-3A (S3A) satellite tracks (tracks 522 and 279), while the red marker indicates the location of the nearby OMNI buoy (BD10). Although the buoy is situated approximately half a degree away from the exact

track intersection, this separation is well within the typical mesoscale decorrelation scales of the region, making it a highly suitable proxy for validating the altimetry-derived currents. The OMNI daily zonal data is selected at the timestamps of S3A mission which has a temporal resolution of 27 days (Supplemental Fig 1a). Figure 2b presents the corresponding time series of the zonal current component (*gu*), its Ekman correction and its validation against the buoy data. The in situ zonal current measured by the OMNI buoy (red line) serves as the reference. When comparing the purely geostrophic zonal current derived from the along-track altimetry (black line) to the buoy data, the altimetry captures the lower-frequency seasonal reversals. To account for the direct wind-driven circulation, the Ekman component was added to the geostrophic flow. As shown by the blue line (geostrophic + Ekman), the inclusion of the Ekman component noticeably improves the agreement with the OMNI buoy observations. The Ekman-corrected timeseries better captures the peak magnitudes (Jul-2019, Aug-2021, Aug-2022) and the higher-frequency variability of the true surface flow. This aligns with the fundamental oceanographic consensus that near-surface buoy measurements represent the total flow (geostrophic plus ageostrophic Ekman currents). Similar improvements in correlation and Root Mean Square Error (RMSE) upon adding the Ekman component have been extensively documented in previous coastal altimetry studies. For instance, Liu et al. (2012) and Saraceno et al. (2008) demonstrated that Ekman-corrected geostrophic currents yield significantly better statistical agreements with moored ADCPs and HF radar observations over continental shelves. Furthermore, global-scale studies by Sudre et al. (2013) and Rio et al. (2014) highlight that capturing the synergistic effect of both components is strictly necessary to accurately retrieve surface layer kinematics. Finally, the time series also includes the zonal current derived from the standard AVISO gridded mapped sea level anomalies (MSLA) at the same intersection point (cyan line). This timeseries closely follows that of track derived geostrophic velocity as is expected in the open ocean. Compared to the along-track altimetry estimates, the AVISO currents exhibit a heavily smoothed temporal profile, failing to capture the sharp, mesoscale peaks recorded by the buoy. This dampening effect is a well-known artifact of the spatial and temporal interpolation schemes used to create gridded altimetry products, which tend to smooth out energetic, small-scale coastal and shelf dynamics (Jebri et al., 2016; Bouffard et al., 2010). Consequently, the along-track formulation combined with the local Ekman correction provides a much more faithful representation of the true surface circulation in this region. At this intersection point, raw track derived geostrophic current gives a correlation of 0.66 against the OMNI, while Ekman corrected timeseries gives a correlation of 0.72.

Building upon the single-point time-series analysis, a comprehensive statistical validation was conducted across the broader Bay of Bengal and Arabian Sea regions in Indian Ocean to robustly quantify the performance of the altimetry-

derived currents. For this basin-scale assessment, co-located measurements from the extensive network of RAMA (Research Moored Array for African-Asian-Australian Monsoon Analysis and Prediction) and OMNI buoys were utilized. As discussed in the data and methods section, zonal and meridional current components can only be resolved at the intersection of reference tracks. Each observation point is surrounded by at least 4 close intersection points (forming the vertices of a diamond shape). The observational timeseries is matched with the geostrophic timeseries of several nearby intersection points after masking the invalid values. This facilitates the estimation of robust comparison metrics between altimetry derived currents and observational currents. Intersection points that are farther than 50km from the observational point are omitted while computing the metrics.

Figure 3 presents density scatter plots comparing the altimetry-derived zonal surface currents against the in situ buoy measurements. To evaluate the specific contribution of wind-driven dynamics, the comparisons are separated into purely geostrophic currents and total currents (geostrophic plus Ekman). Panels (a) and (c) illustrate the comparison using only the purely geostrophic zonal component (u_g) derived from the satellite tracks against the RAMA and OMNI buoys, respectively. While the highest density of points clusters along the theoretical perfect fit (white dashed line), the linear regression (red line) reveals an underestimation of the current magnitudes, and the scatter exhibits considerable spread. Panels (b) and (d) display the same comparisons after the zonal veloctity estimates have been corrected by adding the locally derived zonal Ekman velocity component (u_g+u_E). The inclusion of the ageostrophic Ekman flow yields a clear statistical improvement across both buoy networks:

RAMA Buoys: The number of points available for comparison is quite low (only 205) since all of the equatorial RAMA buoys are omitted (because geostrophy breaks here) with only five buoys remaining for comparison. The wider scatter reflects the inherent challenges of coastal altimetry and the presence of significant ageostrophic processes. The addition of the Ekman component however shifts the linear fit noticeably closer to the 1:1 reference line. The overall correlation (R) improves significantly (from 0.48 to 0.66), and the scatter tightens, reflecting the strong wind-driven dynamics in this region. The standard error has also reduced. Further differences can be accounted for the fact that RAMA bouys' velocities are at 10m and track derived current are surface and almost all the intersection points considered don't fall exactly on the RAMA locations.

OMNI Buoys: OMNI buoy network is situated in the interior of the Bay of Bengal and Arabian Sea. Because OMNI buoys are clustered north of 5N, there are so many spatial and temporal matches with altimetry timeseries. This provided the bulk of points (1148) for the comparison. The high density of points near the origin

indicates that the background state is well-captured, while the tighter clustering along the regression line confirms the high fidelity of the altimeter in these regions. A similar significant improvement is observed for the OMNI network with the addition of Ekman components. The correlation increases (from 0.51 to 0.58), and the standard error is reduced. Across both buoy networks, the integration of the wind-driven Ekman component consistently enhances the accuracy of the satellite-derived surface currents. While Ekman correction failed to reduce RMSE significantly, an additional 7% variance is explained by addition of Ekman drifts.

These statistical improvements align directly with classical physical oceanography principles; near-surface in situ instruments (such as buoys drogued at 10–15 m depth) inherently measure the total kinematic flow, comprising both the pressure-gradient-driven geostrophic currents and the direct wind-driven Ekman drift. On a global scale, studies by Sudre et al. (2013) and Rio et al. (2014) demonstrated that synergistic surface current products combining altimetry with an empirical Ekman model consistently outperform purely geostrophic estimates when validated against Lagrangian drifters and moored Acoustic Doppler Current Profilers (ADCPs). In the context of tropical and equatorial dynamics, Lagerloef et al. (1999) showed that incorporating the Ekman component is strictly necessary to resolve the total velocity variance observed by buoys. Furthermore, regional coastal altimetry assessments, such as those by Liu et al. (2012) on the West Florida Shelf, have consistently reported measurable reductions in Root Mean Square Error (RMSE) and improved correlation coefficients once the wind-driven Ekman velocities are superimposed onto the along-track geostrophic anomalies.

3.2a Near shore validations with Lagrangian drifters

While there are plenty of observations in the open ocean, there is a dearth of velocity observations very close to the coast in the Indian Ocean to make valid comparisons. The velocity data from the ADCP moorings available on the shelf and slope of the east and west coast of India are not valid above 48m (Amol et al 2012, 2014, Mukherjee et al 2014). To further extend validation into the dynamic near-shore domain, we resorted to Lagrangian surface drifters. Drifters provide an excellent source of continuous, high-resolution independent velocity measurements, particularly in regions where moored arrays are sparse. For this assessment, we utilized quality-controlled drifter trajectories that intersected the satellite tracks within a 100 km coastal band. Hundreds of drifter buoys pass close to the Indian coast whose time periods might overlap with the time period of the satellite missions. The spatial and temporal collocation process is done in three steps. First drifter tracks that don't have any overlapping time period with that of the satellite mission are omitted. Then at each intersection point, a small square box of one degree width is formed and it is checked, at each drifter track if any part of it passes through the box. If so, it is added to the candidate list for further processing.

The index of the closest point of the drifter track to the intersection point is computed by least square distance. This index is then used to retrieve the time stamp and zonal velocity of the drifter track. *gu* timeseries is interpolated at the time stamp to get the collocated track zonal velocity. We have chosen the multi-mission reference orbit TPN+J1N+J2N+J3N for this exercise since it has very temporal coverage and 10day temporal resolution, so tolerance of interpolation won't be greater than five days. To focus specifically on the coastal regime, of all the collocated points, only the points that are within 100km distance from the coast are considered for evaluation.

Figure 4 illustrates the density scatter plots comparing the zonal velocity components from the drifter observations against the collocated altimetry-derived estimates for the multi-mission reference orbit (TP+J1+J2+J3+S6A). a) shows the scatter plot comparing the *in situ* drifter zonal velocities (y-axis) against the purely geostrophic zonal currents derived from the along-track sea level anomalies (x-axis). The color bar indicates the point density, revealing that while the highest concentration of measurements clusters along the 1:1 perfect fit reference (white dashed line), there is considerable spread. The linear regression fit (red line) indicates a general underestimation of the current magnitudes by the purely geostrophic altimetry, yielding a baseline correlation and standard error typical of uncorrected satellite estimates in high-variability coastal regions. b) displays the same comparison after the altimetry-derived geostrophic velocities have been corrected by superimposing the locally derived Ekman zonal velocities u_g+u_E. The inclusion of the direct wind-driven component yields a clear statistical improvement. The dense cluster of scatter points aligns much more closely with the 1:1 reference line, the overall correlation increases (from 0.66 to 0.72) and the standard error is visibly reduced (as detailed in the upper left-hand statistics boxes). There is also a modest reduction of RMSE from 0.36m/s(before Ekman correction) to 0.34m/s after correction. Geostrophic current alone explains 48% (R^2 in percentage) and Geostrophic current+Ekman component together explains 56% of total drifter velocity variance. This indicates an additional 8% variance is explained by Ekman component alone in the coastal regions(<100km). It can be noticed that this correlation is better than that of the open ocean. This can be due to the richer spatial temporal availability of drifter data, whereas in open ocean observations are few and fixed. These correlations are similar in magnitude to the geostrophic drifter velocities(after removing the Ekman component) versus altimetry velocities (Niiler et al. 2003). Consistent with the moored buoy validations, the drifter comparisons confirm that near-shore altimetry products successfully capture the large-scale coastal velocity variance.

In the context of boundary currents and coastal regimes, studies by Bouffard et al. (2010) and Liu et al. (2012) have highlighted that coastal altimetry (such as X-

TRACK) can accurately resolve near-shore geostrophic flow, but optimal agreement with *in situ* data (whether from ADCPs, HF radars, or drifters) is only achieved when the wind-driven Ekman velocities are appropriately quantified and added to the signal. Furthermore, Lagerloef et al. (1999) noted that in tropical regimes, incorporating Ekman components is critical for capturing the full variance of drifter velocities. Our results along the Indian coast perfectly mirror these conclusions, demonstrating that the synergistic combination of high-resolution coastal altimetry and Ekman dynamics provides a robust representation of the near-shore circulation.

3.2b Near shore validations with radar

Following the Lagrangian validation using surface drifters, we extend our analysis to an Eulerian framework to evaluate the cross-shelf structure of the East India Coastal Current (EICC). While drifters provide excellent ground truth for tracking water parcels, High-Frequency (HF) radar networks offer a unique advantage by resolving the continuous spatiotemporal evolution of surface currents near the coast. The three radar regions situated in the region have varying temporal variability. Among the three radar regions studied, TN shows better data availability for much of the region. AP radar region shows lowest data availability with less than 50 percent temporal coverage for the whole region. Peripheral regions away from the radar transmitter and receiver locations witness poor data availability for all three regions.

Ekman corrected along-track geostrophic cross-track currents are compared with rotated (in the cross track direction) radar fields extracted along the traversing tracks. Fig 5a shows one such example of traversing track (track number 165 of S3A mission) through OR radar region. Because this descending satellite track is oriented approximately orthogonal to the coastline, the cross-track velocity component serves as an excellent proxy for the along-shelf flow of the EICC.

The time–distance Hovmöller diagrams present the evolution of the along-shelf currents as captured by both the corrected altimetry (Figure 5b) and the HF radar (Figure 5c). b) and c) time versus distance plots of Ekman corrected cross track geostrophic current and radar field extracted along this track. A quantitative assessment of this agreement is detailed in Figure 5d, which illustrates the skill metrics (correlation and Root Mean Square Error, RMSE) as a function of offshore distance (x). The visual agreement between the u_g+u_E structure and the radar-derived velocities is striking. Both datasets successfully capture the pronounced seasonal reversals characteristic of the EICC, transitioning from strong poleward flow in the spring to equatorward flow during the winter monsoon. This robust

seasonality aligns with the acoustic Doppler current profiler (ADCP) observations of the EICC documented by Mukherjee et al. (2014) and the broader boundary current structure modeled by Durand et al. (2009).

Ekman correction has improved noticeably both the correlation and RMSE all across the region. But the overall reduction in RMSE across all transects and all radar regions is negligible (stayed at 0.3m/s). Improvement in correlation is also modest (0.55 mean correlation changed to 0.60). Mean correlations across all traversing tracks is better for regions AP and OR than with TN although TN has better temporal data coverage for most of its region. In radar regions, Ekman correction can only explain an additional variance of 4%. This modest improvement in correlation close to the coast was also noticed in drifter comparisons. However radar metrics are for rotated (in the direction of *gc*) currents, not for zonal currents, so these are not directly comparable.

4. EICC strength along the coast

The East India Coastal Current (EICC) is a western boundary current of the Bay of Bengal. Its features include seasonal reversals driven by monsoon winds and complex spatiotemporal variability (Durand et al., 2009, Shankar 1994, 1996). Flow is dominated by seasonal cycle while intra-seasonal and inter annual variations are highly discontinuous. It flows northward during March to May and reverses southward from September to November. The boundary current is confined within 200km of the east coast of India.

Building upon the robust validation of the Ekman-corrected coastal altimetry currents against independent near-shore radar and drifter observations, we leverage this enhanced dataset to investigate long-term coastal circulation dynamics. Coastal boundary regions historically suffer from a severe scarcity of sustained, multi-decadal in situ current measurements, limiting our understanding of long-term climatic shifts. To address this, we utilize the extensive temporal (from 1992 to 2016) and spatial coverage (788 intersection points, see Sup Fig 1) of the multi-mission reference orbit (comprising ERS-1/2, Envisat, and SARAL/AltiKa) to assess the multidecadal variability of the East India Coastal Current (EICC). Figure 6 illustrates the spatial framework and the resulting long-term timeseries analysis of the EICC's strength. As shown in the left panel, we selected seven altimetry crossover points (labeled A through G, spanning from south to north between 10°N and 22°N) strategically located over the continental slope along the eastern boundary of the Bay of Bengal, primarily bracketed by the 200 m and 1000 m isobaths. Although there are several points on the shelf region very close to the coast, we made sure not to include them since seasonally reversing EICC is absent on the shelf region (Biswmoy 2021). At each of these intersection points, the total surface current speed was computed from the combined geostrophic and Ekman

velocity components. To isolate the long-term multidecadal signal from the highly energetic seasonal monsoonal reversals and interannual perturbations (such as El Niño–Southern Oscillation and Indian Ocean Dipole events), a 500-day low-pass Butterworth filter was applied to the current speed time series. The right panel of Figure 6 presents these low-pass filtered time series alongside their linear trend fits (black dashed lines). The data reveal a pronounced and statistically significant multidecadal weakening of the EICC's strength over the nearly three-decade observation period. The linear trends are predominantly negative across the basin's boundary. Points E, F show the smallest weakening trend while A, B, D points show the strongest trend. As annotated on the spatial map, the reductions in current speed are substantial, reaching up to -0.0223 m s^{-1} yr^{-1} in the southern segment (Point B), with statistically significant weakening extending northward along the coast. This accounts to 55% weakening in the strength of EICC over the last three decades at this point. Averaged across all the points, a 30% weakening of the EICC strength is noticed. While there is considerable decadal variability superimposed on these records, the overall trajectory points toward a systematic deceleration of the boundary current. The significance of this multidecadal weakening within the context of our study is twofold. First, it demonstrates the unique capability of the reprocessed, Ekman-corrected coastal altimetry dataset to capture low-frequency, long-term climate signals right up to the continental slope, a domain where standard gridded altimetry products traditionally fail. Second, it highlights a critical shift in the physical oceanography of the Bay of Bengal. The EICC is a complex western boundary current primarily driven by a combination of local alongshore wind stress, interior Ekman pumping over the Bay of Bengal, and remote equatorial forcing transmitted via coastal Kelvin waves (McCreary et al., 1996; Shankar et al., 1996). A sustained multidecadal weakening suggests a fundamental alteration in one or more of these driving mechanisms.

Driving Mechanisms of the Multidecadal EICC Weakening:

Having identified a robust multidecadal deceleration in the strength of the East India Coastal Current (EICC), it is imperative to investigate the large-scale forcing mechanisms responsible for this trend. The EICC is a dynamically complex boundary current; analytical and numerical modeling studies (e.g., McCreary et al., 1996; Shankar et al., 1996) have established that its seasonal and low-frequency variability is not solely locally driven. Instead, it is governed by a combination of three primary forcings: (1) interior Ekman pumping over the Bay of Bengal, which generates westward-propagating baroclinic Rossby waves; (2) local alongshore wind stress along the east coast of India; and (3) remote equatorial Indian Ocean wind stress, which excites equatorial Kelvin waves that impinge on the eastern boundary and reflect as coastal trapped Kelvin waves propagating up the perimeter of the Bay. To determine which of these mechanisms is driving the observed multidecadal weakening of the EICC, Figure 7 isolates and analyzes the long-term

variability of these specific forcing regions. The left panel defines the spatial masks used for the area-averaged analysis: the equatorial waveguide (5S–5N, green hatching) representing the remote forcing domain; the interior Bay of Bengal gyre region (blue hatching) representing the Ekman pumping domain; and the western boundary along-shore region (red hatching) representing the local wind forcing domain. The right panels (a–d) display the corresponding area-averaged monthly time series. To maintain consistency with the EICC velocity analysis and to isolate the multidecadal signals from the highly energetic seasonal and interannual (e.g., Indian Ocean Dipole) fluctuations, a 500-day low-pass filter was applied to all time series. Panel (b) illustrates the low-pass filtered Sea Level Anomaly (SLA) over the interior Bay of Bengal. The pronounced positive trend reflects the basin-wide sea-level rise, a well-documented consequence of overall warming in the Indian Ocean (Schott et al., 2009, Srinivasu U, 2017). However, the dynamics of the coastal current are dictated by the horizontal pressure gradients driven by the wind field, which are examined in the remaining panels. Panels (a), (c), and (d) present the low-frequency variability and linear trends of the physical forcing parameters: Ekman pumping in the interior, local alongshore wind stress magnitude, and equatorial wind stress magnitude, respectively. The data reveal a discernible downward trend in the magnitude of both the local alongshore wind stress (Panel c) and the remote equatorial wind stress (Panel d). The significance of this figure to the current study is profound, as it provides the dynamical closure to the observations made in Figure 6. The multidecadal weakening of the EICC is not an isolated coastal phenomenon but rather a direct dynamic response to a basin-wide relaxation of its primary atmospheric drivers. A reduction in the equatorial wind stress directly diminishes the amplitude of the eastward-propagating equatorial Kelvin waves, subsequently leading to weaker coastally trapped waves traveling around the Bay of Bengal. Concurrently, the weakening of the local alongshore winds reduces the direct momentum transfer to the boundary current. These findings corroborate recent climate studies which suggest that asymmetric warming across the Indian Ocean has led to a dampening of the overall monsoonal wind circulation (e.g., Roxy et al., 2015). Consequently, the novel coastal altimetry dataset not only resolves the near-shore current dynamics but successfully captures the boundary current's integrated response to shifting basin-wide climatic forcing.

Because the EICC is the primary western boundary current of the basin, the consequences of a weakening EICC are profound and multifold. The EICC plays a critical role in the mass budget of the Bay of Bengal, particularly during its equatorward phase (post-monsoon to winter), when it exports massive volumes of low-salinity riverine water southward along the eastern coast of India. A weakening of this southward flow would reduce the efficiency of this freshwater export. Consequently, this could lead to a progressive freshening and increased

stratification in the northern Bay, while the southern Bay and the southeastern Arabian Sea might experience higher-than-normal salinities.

During the southwest monsoon, the poleward-flowing EICC acts in concert with local southwesterly winds to drive coastal upwelling, drawing cold, nutrient-rich subsurface waters into the euphotic zone. A weakening of both the EICC and its primary atmospheric drivers implies a reduction in the intensity of this upwelling. This would suppress primary productivity along the coast, fundamentally altering the coastal biogeochemical cycling and potentially impacting regional fisheries that rely on these seasonal nutrient pulses.

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